PM-2041 Postmortem - INC-2041 card authorization outage

Date: 2026-03-12
Incident: INC-2041
Written by: Core Banking team
Review status: Reviewed in the weekly operations meeting

Summary

On 2026-03-05 card authorizations at Northwind Payments degraded for 3 h 40 min, from 09:40 to 13:20 UTC. Authorization requests queued inside payments-gateway while waiting for database connections from ledger-service, and a large share of them timed out. This postmortem describes what happened, why it happened and what we are changing. It follows the blameless format: the goal is to improve the system, not to find a person at fault.

The short version: the ledger DB connection pool was too small for normal peak traffic after CHG-0311 reduced it from 200 to 50 connections two days earlier. When the morning traffic peak arrived the pool ran out of free connections, requests waited, and the gateway gave up on them.

Impact

For 3 h 40 min a significant share of card authorization requests failed or were slow. Merchants using our card acceptance products saw declines that were not caused by the cardholder or by fraud rules. Retail merchants with high morning volume were hit hardest. Settlement, payouts and notifications were not affected. No financial data was lost or corrupted; every authorization that completed was recorded correctly in the ledger.

Customer support received a large number of merchant contacts during the morning. Several large merchants asked for a written explanation, which this document and the incident report provide.

Timeline (all times UTC, 2026-03-05)

- 08:30 Morning traffic starts its usual climb as European retail opens.
- 09:00 Scheduled deployment of fraud-scorer model v4 (CHG-0314) completes.
- 09:40 First alerts: payments-gateway request timeouts above threshold. Incident opened.
- 09:52 On-call engineer for payments-gateway joins; gateway pods look healthy, no errors in deploys.
- 10:10 ledger-service dashboards show request queueing; database CPU is normal.
- 10:30 Core Banking on-call confirms the ledger DB connection pool is at 50 of 50 connections in use.
- 10:45 Discussion of raising the pool size; decided to wait for a config review because of CHG-0311.
- 11:15 Incident raised to SEV-1; incident commander assigned.
- 11:40 Gateway timeout rate stable but high; customer support starts merchant communication.
- 12:20 Config review of the pool change starts; rollback of CHG-0311 prepared.
- 13:05 fraud-scorer switched back to model v3 as a precaution while the pool rollback was being prepared.
- 13:20 Connection pool usage drops below the limit, timeouts stop, incident resolved.
- 15:00 Monitoring confirms normal behaviour; incident closed.

Root cause

The root cause was CHG-0311, the reduction of the ledger DB connection pool from 200 to 50 connections

on 2026-03-03. The change was planned as a cost reduction and was based on February monitoring data that showed a typical peak of about 40 open connections. That data used five-minute averages, which hide short bursts. Real peaks during the morning traffic ramp are much higher than the averaged numbers suggest.

Every card authorization takes a ledger DB connection for the duration of the request. When the pool is exhausted, new requests wait in the ledger-service client queue. Waiting requests count against the payments-gateway timeout, so once the queue grew the gateway started to time out requests that would have succeeded with a free connection. Timed-out requests were often retried by merchant terminals, which added more load and kept the pool saturated.

On 2026-03-03 and 2026-03-04 the traffic was lower than on 2026-03-05, which was the first busy weekday morning after the change. That explains why the problem appeared two days after the change rather than immediately.

Contributing factors

- The capacity estimate for CHG-0311 used averaged metrics and did not look at peak concurrency.
- There was no alert on ledger DB pool saturation, so the first signal was gateway timeouts.
- The change was rated low risk, so it was not scheduled with extra monitoring.
- Merchant terminal retries amplified the load once timeouts began.
- fraud-scorer was not a contributing factor. Its model change happened the same morning, but the pool limit would have been reached by normal peak traffic regardless.

Why the change looked safe

The change request for CHG-0311 was reviewed in the normal way and nobody objected. The numbers in the request were correct as averages: over February the pool rarely had more than about 40 connections open in any five-minute window. What the request did not show was the distribution inside those windows. During the morning ramp the number of connections in use rises and falls within seconds, and the short peaks are several times higher than the five-minute average.

The rollback plan in the change request was sound and would have taken about ten minutes. The plan was not used during the incident because the team first wanted to confirm that the pool was the problem and not a symptom of something else. In hindsight, rolling back a recent capacity reduction is cheap and safe, and it should have been the first step.

Detection and response

Detection worked: the gateway timeout alert fired at 09:40, within minutes of the first failed authorizations. Diagnosis was slower. The gateway on-call engineer looked at the gateway first, found healthy pods and no recent gateway deployments, and then looked at downstream services one by one. The ledger-service dashboards showed queueing at 10:10, and the saturated pool was confirmed at 10:30. Because there was no alert on pool usage, this step depended on someone knowing where to look.

Communication to merchants started at 11:40 through customer support. The incident channel had good participation once the incident was raised to SEV-1, but before that only two teams were involved. We will page all critical-path owners earlier in future incidents.

What went well

- Alerting on gateway timeouts fired within minutes of the first failures.
- No authorization data was lost and the ledger stayed consistent throughout.
- Customer support had a holding statement for merchants within the first hour.
- The team that owns ledger-service identified the saturated pool quickly once involved.

What went wrong

- It took 50 minutes from the first alert to look at the ledger DB pool.
- The decision to raise the pool size was delayed by a config review.
- Ownership of the pool configuration was unclear between the gateway and ledger teams.

Lessons learned

Capacity changes to shared resources such as database connection pools need to be checked against peak concurrency, not averages. The ledger DB is shared by the authorization path and by nightly jobs, so its pool is a critical shared resource and should be treated like one. A low-risk rating should not be given to a change that removes three quarters of the capacity of a shared resource.

We also learned that our incident process should bring in the owners of every service on the critical path early. In this incident the ledger-service on-call joined almost an hour after the first alert. Paging all critical-path owners at SEV-2 and above would have saved time.

Action items

- Restore the ledger DB connection pool to 200 connections (CHG-0331). Owner: Core Banking team.
- Add alerts on ledger DB pool saturation at 80% and 95% usage. Owner: Core Banking team.
- Require peak-concurrency data in change requests that reduce shared capacity. Owner: change board.
- Add model load testing with production-like traffic to the fraud model release checklist. Owner: model release process.
- Page all critical-path service owners for SEV-2 and above. Owner: incident management.